# IGCSE 0607 Practice Questions — E10.2

## Interpreting statistical data

**Total Questions:** 6 — **Total Marks:** 34 Marks

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**1** The number of goals scored by Team A in each of their last 6 matches is recorded below.

3 1 0 4 2 2

Team B also played 6 matches.
For Team B, the median number of goals scored was 1.5 and the range was 5.

**(a)** Work out the median and the range of the number of goals scored by Team A.

Median = ..... , Range = ..... **[2]**

**(b)** Make two comments comparing the performance of Team A with the performance of Team B.

.....

.....

..... **[2]**

**2** The table shows information about the mass, $m$ grams, of 60 mangoes collected by a farmer.

| **Mass ($m$ grams)** | $100 < m \leq 120$ | $120 < m \leq 140$ | $140 < m \leq 160$ | $160 < m \leq 180$ | $180 < m \leq 200$ |
|---|---|---|---|---|---|
| **Frequency** | 10 | 20 | 15 | 10 | 5 |

**(a)** State the modal class.

..... **[1]**

**(b)** Calculate an estimate of the mean mass of the mangoes.

..... $g$ **[4]**

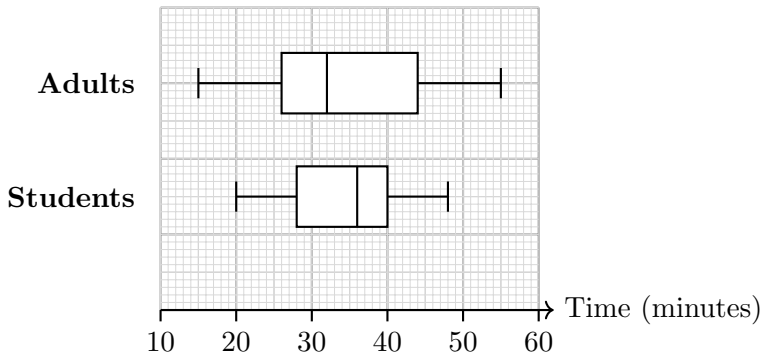
**(c)** The farmer states that exactly 15 mangoes have a mass of 150 grams because 150 is the midpoint of the class $140 < m \leq 160$.
Explain why the farmer's reasoning is incorrect.

.....

.....

..... **[2]**

**3** The box-and-whisker plots show information about the times, in minutes, taken by a group of Adults and a group of Students to complete a logic puzzle.




**(a)** Use the diagram to find the interquartile range of the times taken by the Adults.

..... minutes **[2]**

**(b)** Make two different statements comparing the distribution of the times taken by the Adults with the times taken by the Students.

.....

.....

..... **[2]**

**(c)** A student claims: "Because the range of the Students' times is smaller, this proves that every student completed the puzzle faster than the slowest Adult."

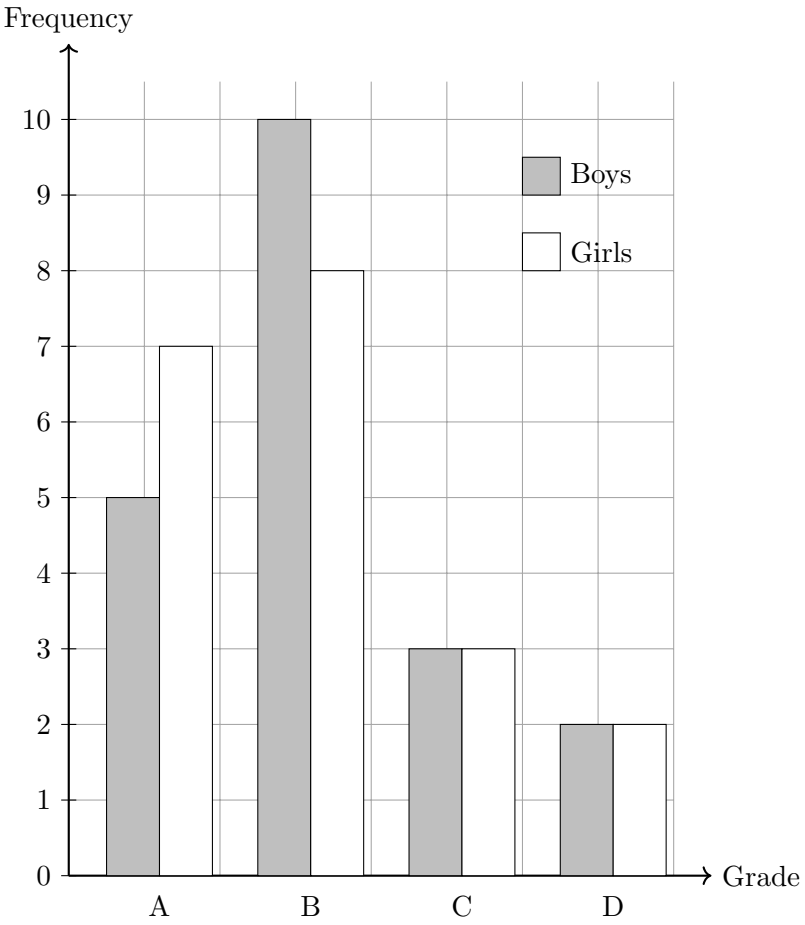
Comment on the validity of this reasoning.

.....

.....

..... **[2]**

**4** The dual bar chart shows the grades achieved by a group of boys and a group of girls in an examination.




**(a)** Write down the modal grade for the boys.

..... **[1]**

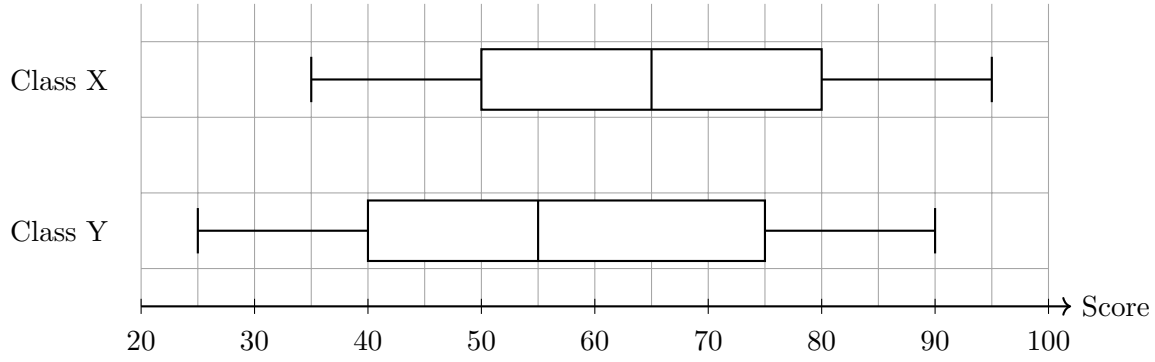
**(b)** Find the number of girls who scored Grade C.

..... **[1]**

**(c)** Calculate the percentage of all students who scored Grade A.

..... % **[2]**

**5** The box-and-whisker plots show the distribution of test scores for two classes, Class X and Class Y.




**(a)** Write down the median score for Class X.

..... **[1]**

**(b)** Work out the interquartile range for Class Y.

..... **[2]**

**(c)** Make two different comments comparing the distribution of test scores in Class X with Class Y.

**[2]**

**(d)** A student states: "Because the median score for Class X is higher, every student in Class X scored more than every student in Class Y."

Is this statement correct?
Give a statistical reason for your answer.

.....

..... **[1]**

**6** A set of 5 integers has the following statistical properties:

- Mode = 3
- Median = 7
- Mean = 8
- Range = 11

**(a)** Find the 5 integers.

..... **[4]**

**(b)** A sixth integer, $x$, is added to the set.
The new mean is 9.
Work out the value of $x$.

$x =$ ..... **[2]**

**(c)** A student claims: "If another completely different set of 5 integers has the same range of 11, it must also have the same standard deviation."

State whether the student is correct.
Give a statistical reason for your answer.

.....

..... **[1]**